

Java Fullstack Roadmap: From Core to Spring Boot & Hibernate

Basics of Java

- Introduction to Java
- Installing JDK & IDE
- Hello World Program
- Java Syntax and Structure
- Data Types
- Variables and Constants
- Operators
- Control Flow (if, switch)
- Loops (for, while, do-while)
- Methods (Functions)
- Method Overloading
- Arrays
- String Handling
- User Input (Scanner)
- Basic OOP Concepts

Object-Oriented Programming (OOP)

- Classes and Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Interfaces
- Abstract Classes
- this and super keywords
- Static Members

Intermediate Java

- Exception Handling

- File I/O (java.io)
- Collections Framework
- List, Set, Map
- Generics
- Wrapper Classes
- Enum Types
- Varargs
- Java Annotations
- Java Packages and Imports

Advanced Java

- Multithreading
- Thread Lifecycle
- Synchronization
- Java Concurrency (Executors, Callable)
- Lambda Expressions
- Streams API
- Functional Interfaces
- Date and Time API (java.time)
- Reflection API
- JVM Internals & Garbage Collection

Java Development Tools

- Maven
- Gradle
- JUnit Testing
- Log4j/SLF4J Logging
- JavaDoc
- Build Lifecycle
- Dependency Management

JDBC & Database Connectivity

- What is JDBC?
- JDBC Architecture

- Connecting to Databases
- Executing SQL Queries
- PreparedStatement vs Statement
- ResultSet
- Connection Pooling
- Transaction Management

Spring Boot Framework

- What is Spring Boot?
- Spring Boot Starter Projects
- Spring Boot Annotations
- Creating REST APIs
- Dependency Injection
- Spring Boot DevTools
- Spring Boot Properties
- Exception Handling in Spring Boot
- Spring Boot Testing
- Spring Boot Auto Configuration

Spring Data JPA & Hibernate

- What is JPA?
- Spring Data JPA Overview
- Repositories (CrudRepository, JpaRepository)
- Entity Relationships (OneToMany,ManyToOne, etc.)
- JPQL vs Native Queries
- Pagination and Sorting
- Entity Lifecycle
- What is Hibernate?
- Hibernate Caching
- Hibernate Validator

Java Web Development

- Servlets and JSP
- MVC Architecture

- RESTful Services
- Spring MVC
- Thymeleaf
- Handling JSON/XML in APIs
- Web Security Basics
- Spring Security Intro
- Authentication and Authorization
- JWT with Spring Boot

Deployment and Cloud

- Packaging Java Applications
- Deploying on Tomcat
- Dockerizing Java Apps
- Spring Boot with Docker
- Using PostgreSQL/MySQL in Production
- Cloud Deployment (Heroku, AWS)
- CI/CD Concepts
- Monitoring with Actuator
- Logging with ELK
- Environment Profiles